

A growing epithelial spheroid in an extracellular matrix: the Plexus2 implementation

prototype/ecm · runs in log/okuda_ECM · 6 August 2026

1. Sets, operators, schedule

A Plexus model is sets $\times$ fields $\times$ operators $\times$ schedule. The biological hierarchy here is four entities and one field; the basement membrane is a *specialised* ECM, not the ECM:

entity	set	representation	principal operators
epithelium	vertex , cell	3D active vertex model	shape_energy_3d , morphogen_growth_3d , d
junction	relation on the mesh	half-edges, keyed state	junction_myosin , reconnect_t1_3d
basement membrane	basement_membrane_particle	MPM + explicit bonds	.._seed/bond/remodel/bond_break , integr
interstitial ECM	mpm_particle	MPM fibres	ecm_seed , ecm_stress , cell_to_ecm
(field)	mpm_grid	MLS-MPM background grid	mpm_scatter/gather/grid_update

Every particle body scatters into and gathers from the *same* grid; that sharing *is* the mechanical coupling between them, so no pairwise contact model is written. Operators carry one of seven kinds. The four dynamics kinds return a per-set delta the engine integrates once per tick; **field**, **rewire** and **structural** mutate the grid, the relation E , or the membership $|S|$ in place and *return nothing*. That last fact drives a design decision in §5.

2. The spheroid: 3D active vertex model

Cells are faces of a closed triangulated shell; cell f 's volume is the wedge from the tissue centre, $v_f = \frac{1}{3} \mathbf{c}_f \cdot \mathbf{N}_f$. The energy is

$$E = \sum_f \left[K_A (A_f - A_f^0)^2 + K_P (P_f - P_f^0)^2 + \frac{1}{2} \Gamma P_f^2 + K_V (v_f - v_f^0)^2 \right] + \Lambda \sum_e m_e \ell_e + K_R \sum_i (|\mathbf{x}_i| - R_0)^2, \quad (1)$$

relaxed overdamped each frame, $\mathbf{x} \leftarrow \mathbf{x} - \eta \mu \nabla E$ with a per-step cap. The factor m_e is new (§5); with $m_e \equiv 1$ Eq. (1) is the stock Okuda energy and every prior run is bit-identical. Growth scales each cell's targets by a cumulative s_f ,

$$s_f \leftarrow s_f (1 + \lambda(\rho + \text{Hill}(a_f))), \quad A_f^0 = A_f^{0,\text{init}} s_f^2, \quad P_f^0 = P_f^{0,\text{init}} s_f, \quad v_f^0 = v_f^{0,\text{init}} s_f^3, \quad (2)$$

and **divide_3d** splits a cell once v_f passes twice its reference. From 200 cells the tissue reaches ~ 6000 and its apical radius $4.66 \rightarrow 16.6$ (arbitrary units) in 402 frames.

3. The ECM: MLS-MPM

The matrix is material points on a background grid, fixed-corotated with $F = RS$:

$$\boldsymbol{\tau} = 2\mu (F - R)F^\top + \lambda J(J - 1)I, \quad \boldsymbol{\sigma} = \boldsymbol{\tau}/J, \quad \sigma_{\text{vM}} = \sqrt{\frac{3}{2} \mathbf{s} : \mathbf{s}}, \quad \mathbf{s} = \boldsymbol{\sigma} - \frac{1}{3} \text{tr}(\boldsymbol{\sigma})I. \quad (3)$$

$\boldsymbol{\tau}$ was already computed every substep to build the affine momentum matrix and then discarded; **mpm_scatter** now optionally caches the Cauchy stress $\boldsymbol{\sigma}$ to a per-particle buffer (**store_stress**, off by default, byte-identical when off). This matters quantitatively: colouring by the volumetric measure $|J - 1|$ put 0.6% of the matrix above the display floor, whereas σ_{vM} of the *same* run puts 64% there. A matrix pushed by a growing sphere *shears*; the fixed-corotated law resists volume change far more stiffly than shape change, so the volumetric measure was blind to the deformation, not the mechanics.

4. Spheroid $\leftrightarrow$ ECM

The two solvers cannot share a world (**mpm_grid** is unit-box; the vertex model lives in a 50-unit box and collapses if rescaled), so they are run as two passes joined by one geometric scale. The tissue surface is recorded as an angular radius map $R(\theta, \phi, t)$ over its apical vertices. Contact is a one-sided penalty plus a hard projection,

$$\mathbf{a}_i = k (R(\hat{\mathbf{u}}_i) - r_i) \hat{\mathbf{u}}_i \text{ for } r_i < R, \quad \text{then} \quad \mathbf{x}_i \leftarrow \mathbf{c} + \hat{\mathbf{u}}_i R(1 + \varepsilon) \text{ if still inside,} \quad (4)$$

the projection (**cell_exclude_3d**) being necessary because a penalty punishes penetration rather than preventing it. The reaction is accumulated by direction into a pressure map and normalised by the solid angle each bin subtends, $P(\theta, \phi, t) = \sum_{i \in \text{bin}} k \text{depth}_i / (R^2 d\Omega)$, which closes the loop back onto the cells: **ecm_growth_gate_3d** slows the cell cycle where the matrix presses,

$$g = f_0 + \frac{1 - f_0}{1 + (P/P_{1/2})^n}, \quad v_f^0 \leftarrow v_f^{0,\text{prev}} + g (v_f^{0,\text{now}} - v_f^{0,\text{prev}}), \quad (5)$$

i.e. the growth *increment* of Eq. (2) is gated, so a loaded cell reaches its division threshold later and divides less often. This is a rate effect and compounds: it turns a $2.75\times$ stress anisotropy into a shape. Measured, against a round control at oblateness 1.009: gating on a polar stress pattern gives 1.32–1.43, while a *flat* pattern of the same strength shrinks the tissue just as hard and leaves it round (1.015). The shape comes from the pattern, not the amount. The binding constraint proved to be *when* the pattern exists, not how strong it is: five interventions that raised the stress (stiffer matrix, denser matrix, a trapping plane, an earlier gate, plates in contact) each *lowered* the shape, and only an oblate cavity—which makes the pattern directional from first contact, 14.7 against 1.5 at frame 250—reached the ideal.

5. The two new levels

Junction myosin. Actomyosin previously existed only as the scalar Λ of Eq. (1). It is now per-junction, recruited by tension and relaxing over τ_m :

$$m_e^{ss} = \alpha \left(1 + \beta (\ell_e / \langle \ell \rangle - 1) \right), \quad \frac{dm_e}{dt} = \frac{m_e^{ss} - m_e}{\tau_m}, \quad (6)$$

with α the global activity—the *in silico* myosin inhibition. Because **rewire** and **structural** operators return no delta (§1), any state living on junctions is silently invalidated by every T1, division and death. So m_e is keyed by *topological identity*, the unordered vertex pair $(\min(v_i, v_j), \max(v_i, v_j))$: a surviving edge finds its own value, a new edge takes m_{new} , a deleted edge stops being asked, and **reconnect_t1_3d** and **divide_3d** need no edits. A test battery confirms $\alpha=1, \beta=0$ is bit-identical to no operator, that the radius falls monotonically over $\alpha \in [0, 3]$ ($7.22 \rightarrow 6.14$), and that new junctions appear at 10.5 per division rather than at the edge count per frame—the signature that the keying holds.

Basement membrane. A shell of material points seeded on $R(\hat{\mathbf{u}}, 0) + \delta$, with explicit crosslinks, so that it can be *defective*—MPM alone has no connectivity and cannot fragment. Three terms, plus adhesion:

$$\text{crosslink: } \mathbf{f}_{ij} = k_b (\ell_{ij} - \ell_{ij}^0) \hat{\mathbf{d}}_{ij}, \quad \text{turnover: } \ell_{ij}^0 \leftarrow \ell_{ij}^0 + \frac{\ell_{ij} - \ell_{ij}^0}{\tau_r}, \quad \text{failure: } \frac{\ell_{ij}}{\ell_{ij}^0} - 1 > \varepsilon_b \Rightarrow \text{bond removed}, \quad (7)$$

$$\text{integrin: } \ddot{\mathbf{x}}_i = k_a (\mathbf{a}_i(t) - \mathbf{x}_i) - c_d \dot{\mathbf{x}}_i, \quad \mathbf{a}_i(t) = \mathbf{c} + \hat{\mathbf{u}}_i (R(\hat{\mathbf{u}}_i, t) + \delta), \quad c_d = 2\sqrt{k_a} (\zeta = 1). \quad (8)$$

Bond removal is registered **rewire**: fragmentation *is* a change of E . What is reported is not the bond count but the largest connected component—a sheet that has lost a third of its crosslinks and is still one piece is not fragmented.

Two calibration facts, both found by tests rather than by inspection. (i) Writing the crosslink force as $k\varepsilon = k(\ell - \ell^0)/\ell^0$ makes it $1/\ell^0 \approx 450\times$ stiffer than the number supplied, and the sheet tore itself apart in 40 frames; Eq. (7) is Hookean in the *extension*, with the explicit-integration bound $\Delta t_{\text{sub}} < 2/\sqrt{k_b}$. (ii) The anchor in Eq. (8) *moves*, at $v_a \approx 0.135$ box/time. An undamped spring does not track a ramp, it oscillates about it with amplitude $v_a/\omega = 9.5 \times 10^{-4}$, against a bond rest length of 2.2×10^{-3} : 43% strain oscillation on every particle, and 66 680 of 70 129 crosslinks broken. At $\zeta = 1$ the steady lag $2v_a/\omega$ is larger but *coherent* between neighbours, so it does not stretch bonds—0 broken, the sheet intact at 10% strain.

A result falls out of Eq. (7) without τ_r : sweeping ε_b over 0.05/0.20/0.60 destroyed the sheet at every value, because the surface grows 66% in linear strain by frame 150 and 260% by the end. A purely elastic basement membrane *cannot* enclose a growing epithelium; remodelling is not optional, and fragmentation becomes a race between τ_r and growth.

6. Status and the next operator

63 runs in `log/okuda_ECM`; `prototype/ecm/RUNS.md` records which are mutually comparable, since three quantities were redefined mid-campaign (notably $|J - 1| \rightarrow \sigma_{\text{VM}}$, $\sim 100\times$ in “strained fraction”). The coupling of §4 is one-way; `ecm_load_3d` is the staggered two-way scheme, implemented and uncalibrated. `cell_polarity` is absent—the epithelium has no simulated basal side, its cells being wedges from the centre and the monolayer a rendering convention—as are `cell_extrude_3d` and MMP degradation.

The clearest gap is in Eq. (7). With N fixed on a shell whose radius grows $3.6\times$, the areal density $N/4\pi R^2$ falls $13\times$ and every rest length must creep to $3.6\times$ its original span: what τ_r delivers is therefore *dilution with amnesia*, not remodelling, and it is why the surviving sheet sits permanently near 10% strain. A basement membrane does not thin thirteenfold as an acinus grows—the epithelium secretes laminin and collagen IV continuously and thickness is held roughly constant. The missing operator is **basement_membrane_secrete**, kind **structural**: insert particles where areal density has fallen below target and bond them in, the sheet’s analogue of `divide_3d`. Plexus supports it directly through `Level.free_slots/spawn` with an oversized reservoir. Only then does “the membrane survived” mean what it means biologically—that the tissue did not outgrow the membrane it was building—rather than that its crosslinks were stretched and told to forget.